

Topological Integrity Gravity: A Structural Horizon Transition

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Abstract

We analyze the horizon structure of a specified static, spherically symmetric representative geometry within the Topological Integrity Gravity (TIG) research program. The chosen regular mass profile yields the exact dimensionless horizon equation $x^3 - x^2 + \beta^3 = 0$. We derive its discriminant, positive-root sectors, and critical point $x_c = 2/3$, $\beta_c = (4/27)^{1/3}$. The two positive roots coalesce in a nondegenerate local fold with square-root separation. The cubic is a consequence of the chosen mass profile, not a universal requirement of fold bifurcations or dimensional consistency. A direct trace test also excludes this nontrivial representative metric as an exact vacuum solution of constant-coefficient quadratic metric $f(R)$ gravity. This geometry is a reparameterized Hayward metric. The Schwarzschild limit and bounded-curvature core remain geometric properties. A prescribed reflecting model yields a conditional inverse-square-root round-trip delay on the horizonless side. Physical reflection, dynamical completion and observational forecasts remain separate tasks.

1 Introduction

General Relativity provides the most successful classical description of gravitation. Its geometric formulation identifies gravitational interaction with the curvature of spacetime and has passed a wide range of observational tests since Einstein’s formulation of the gravitational field equations [1, 3, 4]. Nevertheless, the appearance of singularities and the structure of horizon formation indicate that the classical theory may not fully encode all structural constraints relevant at extreme curvature scales.

A common strategy is to modify the gravitational action by adding curvature corrections. Such models are often written as extensions of the Einstein-Hilbert action through a generalized function $f(R)$ of the Ricci scalar. These approaches form a broad and well-studied class of modified gravitational theories [5–7]. However, this procedure may introduce an overly large space of possible theories. If curvature corrections are chosen freely, the resulting model space becomes phenomenological rather than structurally constrained.

The purpose of this work is to formulate a minimal alternative viewpoint. We assume that admissible curvature corrections are restricted by consistency conditions. In this sense, the question is not merely which correction terms can be written down, but which correction terms preserve the structural integrity of horizon formation.

We call this framework *Topological Integrity Gravity* (TIG). The term “topological integrity” is used here in a restricted mathematical sense: it refers to the preservation of the qualitative root structure associated with horizon formation under admissible curvature corrections.

The central result of this paper is the exact horizon equation of the chosen representative mass profile,

$$x^3 - x^2 + \beta^3 = 0, \quad (1)$$

where x is a dimensionless radial horizon variable and β is a dimensionless structural control parameter. This equation exhibits a critical transition: for sufficiently small β , horizon solutions exist; beyond a critical value, no positive horizon root remains.

The transition follows from the algebraic horizon equation once the mass profile has been specified. Structural consistency requirements alone do not select that profile uniquely. The resulting critical point is a boundary between horizon and non-horizon configurations within this model. The associated root coalescence is structurally analogous to fold-type criticality and saddle-node bifurcations known from catastrophe theory [10].

The representative geometry below is exactly Hayward’s static family [13] after setting $r_c^3 = 2M\ell^2$. Its metric and horizon coalescence are known properties, not new geometric discoveries. At fixed ℓ , varying M changes r_c ; a fixed- r_c parameter scan is not Hayward’s time evolution.

2 Geometric and Physical Setup

We consider a four-dimensional spacetime (M, g) equipped with a Lorentzian metric $g_{\mu\nu}$. The discussion is restricted to static, spherically symmetric configurations. The general line element may be written in the form

$$ds^2 = -A(r)c^2 dt^2 + B(r)dr^2 + r^2 d\Omega^2, \quad (2)$$

where $d\Omega^2$ denotes the metric on the unit sphere.

In the Schwarzschild solution of General Relativity, the horizon is located at the radial coordinate

$$r_s = \frac{2GM}{c^2}, \quad (3)$$

where M is the mass of the central object [2]. The Schwarzschild geometry remains the canonical reference solution for static black holes in General Relativity [3, 9].

In the one-function static sector used below, candidate Killing horizons are located at positive zeros of the lapse function,

$$A(r) = 0. \quad (4)$$

In modified or structurally constrained theories, the functional form of $A(r)$ may receive corrections. The key question is whether such corrections preserve a well-defined horizon structure.

Definition 1 (Dimensionless horizon variable). *We define the dimensionless radial variable*

$$x := \frac{r}{r_s}. \quad (5)$$

The Schwarzschild horizon corresponds to $x = 1$.

Definition 2 (Structural control parameter). *Let ℓ denote a characteristic correction length scale associated with the regularization or structural modification of the central region. We define the dimensionless parameter*

$$\beta := \frac{\ell}{r_s}. \quad (6)$$

The regime $\beta \ll 1$ corresponds to corrections much smaller than the Schwarzschild scale.

Below we use geometrized units $G = c = 1$, with M denoting the mass in length units, $r_s = 2M$, and $\ell = r_c$. The analyzed sector has $A(r) = F(r)$ and $B(r) = 1/F(r)$; it does not exhaust general static spherical metrics. Counting its lapse zeros does not establish a dynamical formation process or the global event-horizon structure of an evolving spacetime.

The limit $\beta = 0$ must recover the classical Schwarzschild horizon. This requirement imposes a normalization condition on any admissible horizon equation: the dimensionless equation must admit $x = 1$ as the physical root when $\beta = 0$.

3 Structural Consistency Principle

The guiding assumption of TIG is that gravitational corrections are not arbitrary deformations. Instead, they must preserve the qualitative integrity of the horizon formation problem.

Definition 3 (Structural horizon consistency). *A horizon equation is structurally consistent if:*

- (i) it admits the Schwarzschild limit $x = 1$ when the correction parameter vanishes;*
- (ii) it has a well-defined algebraic root structure;*
- (iii) horizon disappearance, if it occurs, takes place through a critical root coalescence rather than by arbitrary removal of solutions;*
- (iv) the correction terms are dimensionless and compatible with the chosen normalization;*
- (v) the resulting transition can be analyzed through invariant algebraic quantities such as the discriminant.*

These conditions are minimal. They do not specify a complete gravitational field theory. Instead, they constrain the admissible reduced horizon equations.

These requirements do not determine a polynomial degree or a unique mass profile. A local fold has an equilibrium equation of the form

$$y^2 - \lambda = 0, \quad (7)$$

which is quadratic; a corresponding potential can be cubic [10, 12]. For example, $Q(x, \lambda) = x^2 - x + \lambda$ has the Schwarzschild-normalized root $x = 1$ at $\lambda = 0$ and a fold at $(x, \lambda) = (1/2, 1/4)$. Adding a double zero-radius factor is an additional model choice, not a consequence of fold criticality. Likewise, dimensional consistency of two dimensionless variables does not require a β^3 correction.

4 Derivation of the Specific Cubic Horizon Equation

Choose $M > 0$, $r_c > 0$, and the representative mass profile

$$m(r) = \frac{Mr^3}{r^3 + r_c^3}, \quad F(r) = 1 - \frac{2m(r)}{r}. \quad (8)$$

It has $m(r) \rightarrow M$ at infinity and $m(r) = O(r^3)$ near the center. These requirements do not make the interpolation unique: for example, $\tilde{m}(r) = Mr^3/(r^2 + r_c^2)^{3/2}$ has the same two limiting properties and gives a different horizon equation.

For the chosen profile and positive horizon radius, $F(r_H) = 0$ gives

$$r_H^3 - 2Mr_H^2 + r_c^3 = 0. \quad (9)$$

With $x = r_H/(2M)$ and $\beta = r_c/(2M)$, division by $(2M)^3$ yields

$$P(x, \beta) = x^3 - x^2 + \beta^3 = 0. \quad (10)$$

Indeed, $F = P/(x^3 + \beta^3)$; the denominator is positive for $x > 0$, $\beta > 0$. The β^3 term comes from the chosen r_c^3 in the mass-profile denominator, not from dimensional analysis alone. This is an exact reduced horizon equation for the specified geometry.

At $\beta = 0$, clearing the denominator yields $P_0 = x^2(x - 1)$. The double zero at $x = 0$ is outside the equivalence domain and is not an additional Schwarzschild horizon or a derived internal physical sector. For $r_c > 0$, the regular center has $F(0) = 1$.

Remark 1. Equation (10) is not a fundamental field equation or a unique consequence of the consistency principle. Its cubic degree is a property of the selected representative geometry.

5 Root Structure and Criticality

The cubic equation

$$x^3 - x^2 + \beta^3 = 0 \quad (11)$$

has a critical point when two real roots coalesce. This occurs when the polynomial and its derivative vanish simultaneously.

The derivative is

$$\frac{dP}{dx} = 3x^2 - 2x = x(3x - 2). \quad (12)$$

The nonzero stationary point is therefore

$$x_c = \frac{2}{3}. \quad (13)$$

At criticality, we impose

$$P(x_c, \beta_c) = 0. \quad (14)$$

Substituting $x_c = 2/3$, we obtain

$$\left(\frac{2}{3}\right)^3 - \left(\frac{2}{3}\right)^2 + \beta_c^3 = 0. \quad (15)$$

This gives

$$\beta_c^3 = \frac{4}{27}, \quad (16)$$

and hence

$$\beta_c = \left(\frac{4}{27}\right)^{1/3}. \quad (17)$$

Proposition 1 (Critical horizon transition). *The TIG horizon equation admits a critical transition at*

$$\beta_c = \left(\frac{4}{27}\right)^{1/3}. \quad (18)$$

At this point, two real roots coalesce at

$$x_c = \frac{2}{3}. \quad (19)$$

Proof. A double root occurs when $P(x, \beta) = 0$ and $P'(x, \beta) = 0$. Since $P'(x) = x(3x - 2)$, the nonzero candidate is $x = 2/3$. Substitution into $P(x, \beta) = 0$ yields $\beta_c^3 = 4/27$, completing the proof. $\square$

6 Discriminant Analysis

The cubic equation

$$x^3 - x^2 + \beta^3 = 0 \quad (20)$$

can be written in the standard form

$$x^3 + ax^2 + bx + c = 0, \quad (21)$$

with

$$a = -1, \quad b = 0, \quad c = \beta^3. \quad (22)$$

The discriminant of a cubic is

$$\Delta = a^2b^2 - 4b^3 - 4a^3c - 27c^2 + 18abc. \quad (23)$$

Substituting $a = -1$, $b = 0$, and $c = \beta^3$, we find

$$\Delta = 4\beta^3 - 27\beta^6. \quad (24)$$

Factoring,

$$\Delta = \beta^3(4 - 27\beta^3). \quad (25)$$

The discriminant vanishes when $\beta = 0$ or when

$$\beta_c^3 = \frac{4}{27}. \quad (26)$$

The sign of the discriminant determines the real root structure:

- $\Delta > 0$: three distinct real roots;
- $\Delta = 0$: multiple roots;
- $\Delta < 0$: one real root and two complex conjugate roots.

The discriminant therefore provides an invariant algebraic diagnostic of the structural transition.

7 Branch Geometry and Physical Interpretation

The positive roots of the horizon equation correspond to candidate horizon locations. In the Schwarzschild limit $\beta = 0$, the equation becomes

$$x^3 - x^2 = x^2(x - 1) = 0. \quad (27)$$

The physical nonzero root is

$$x = 1. \quad (28)$$

For $0 < \beta < \beta_c$, there are exactly two positive roots and one negative root. The positive roots follow from $P(0) > 0$, $P(2/3) = \beta^3 - 4/27 < 0$, and the monotonicity on either side of the minimum. The negative root is outside the radial domain. The outer positive root is continuously deformed away from the Schwarzschild value. As β increases, the relevant positive roots approach each other. At $\beta = \beta_c$, the roots coalesce at $x = 2/3$. For $\beta > \beta_c$, the corresponding pair of real roots no longer exists.

At criticality, $P = (x - 2/3)^2(x + 1/3)$; above criticality its only real root is negative. Thus the count of positive horizon roots changes from two to zero through one positive double root. The branch locations are continuous up to coalescence; it is their number that changes. This is a model-specific static horizon transition, not a proof of dynamical black-hole formation.

8 Comparison with General Relativity

For every fixed $r > 0$, the limit $r_c \rightarrow 0$ of the chosen geometry is $F(r) = 1 - 2M/r$, and its outer horizon tends to $r = 2M$ or $x = 1$. This is a metric-level Schwarzschild recovery, not a derivation of the full Einstein dynamics from a completed TIG action. The limit is not uniform at the center, where the Schwarzschild geometry is singular.

9 Scaling Near Criticality

Near the critical point, root coalescence implies non-linear sensitivity. Let

$$\epsilon = \beta_c - \beta. \quad (29)$$

For small positive ϵ , the separation between the coalescing roots is expected to scale as a fractional power of ϵ .

To see this, expand $P(x, \beta)$ near the critical point:

$$x = x_c + \delta, \quad \beta = \beta_c - \epsilon. \quad (30)$$

Since

$$P(x_c, \beta_c) = 0, \quad P_x(x_c, \beta_c) = 0, \quad (31)$$

the leading expansion is

$$P(x, \beta) \approx \frac{1}{2} P_{xx}(x_c, \beta_c) \delta^2 + P_\beta(x_c, \beta_c) (-\epsilon). \quad (32)$$

The derivatives are

$$P_{xx} = 6x - 2, \quad P_\beta = 3\beta^2. \quad (33)$$

At $x_c = 2/3$, one has $P_{xx}(x_c) = 2$. Hence

$$\delta^2 \sim 3\beta_c^2 \epsilon, \quad (34)$$

and therefore

$$\delta_\pm = \pm\sqrt{3}\beta_c\epsilon^{1/2} + O(\epsilon). \quad (35)$$

Here $P_{xx}(x_c, \beta_c) = 2 \neq 0$ and $P_\beta(x_c, \beta_c) = 3\beta_c^2 \neq 0$, establishing a nondegenerate local fold [10, 12]. The exponent refers to horizon-root displacement in this control parameter; it does not by itself fix an echo, shadow, or perturbation exponent or imply dynamical stability.

10 Representative Effective Geometry

For illustration, one may consider a representative bounded-curvature lapse function of the form

$$F(r) = 1 - \frac{2Mr^2}{r^3 + r_c^3}, \quad (36)$$

where r_c is a regulated core scale. This expression is not introduced as a complete solution of new covariant field equations. It is a representative geometry used to examine the compatibility of the cubic horizon structure with a regularized high-curvature sector.

For large radial distance, $r \gg r_c$, one has

$$F(r) = 1 - \frac{2M}{r} + \frac{2Mr_c^3}{r^4} + O(r^{-7}), \quad (37)$$

so the Schwarzschild-compatible asymptotic behavior is recovered.

Near the origin, the lapse takes the form

$$F(r) = 1 - \frac{2M}{r_c^3}r^2 + O(r^5), \quad (38)$$

which corresponds to a regular de Sitter-like core in this representative sector. This does not constitute a general singularity-resolution theorem; it only states that the analyzed representative geometry has finite central behavior.

10.1 Test Against Quadratic Metric Gravity

For comparison, consider the metric action $S_g = (16\pi)^{-1} \int \sqrt{-g}(R - 2\Lambda + \mu R^2) d^4x$, with constant finite μ and Λ . Its vacuum trace equation is

$$6\mu\Box R - R + 4\Lambda = 0 \quad (39)$$

[6, 11]. For the representative geometry, writing $a = r_c^3$, direct curvature calculation gives

$$R(r) = \frac{12Ma(2a - r^3)}{(r^3 + a)^3}, \quad \Box R = \frac{1}{r^2} \frac{d}{dr} \left(r^2 F \frac{dR}{dr} \right). \quad (40)$$

At infinity the trace residual tends to 4Λ , so vacuum compatibility requires $\Lambda = 0$. With that value,

$$\lim_{r \rightarrow \infty} r^6(6\mu \square R - R) = 12Ma \neq 0, \quad \lim_{r \rightarrow 0^+} (6\mu \square R - R) = -\frac{24M}{a} \neq 0. \quad (41)$$

Consequently, for $M > 0$ and $r_c > 0$, this metric is not an exact vacuum solution of constant-coefficient quadratic metric gravity. With matter, a specified source and its own equations must instead be demonstrated; defining an effective tensor from the metric is insufficient. No relation between r_c and μ is derived here. This negative dynamical test does not alter the algebraic horizon calculation for the representative model.

11 Photon Sphere Structure and Observable Deviation

A central consequence of the TIG framework is the modification of null geodesic structure near compact objects. In particular, the location of the photon sphere provides a direct probe of the underlying gravitational geometry.

The photon sphere is determined by the standard condition for unstable circular null geodesics:

$$rF'(r) - 2F(r) = 0. \quad (42)$$

Within the representative TIG geometry, the metric function takes the form

$$F(r) = 1 - \frac{2Mr^2}{r^3 + r_c^3}, \quad (43)$$

which introduces a characteristic scale r_c governing deviations from the classical Schwarzschild geometry.

In the limit $r_c \rightarrow 0$, the model smoothly reduces to the Schwarzschild solution with

$$r_{\text{ph}}^{\text{GR}} = 3M, \quad (44)$$

ensuring consistency with established gravitational theory.

We define the relative deviation from the Schwarzschild photon sphere as

$$\Delta_{\text{ph}} = \frac{r_{\text{ph}}^{\text{TIG}} - 3M}{3M}. \quad (45)$$

The circular-null-orbit equation specifies how to compute the photon-sphere shift for this representative metric. A verified numerical curve and its parameter domain are not supplied in this manuscript; the equation alone does not establish an observational forecast.

A quantitative shadow calculation additionally requires the admissible orbit branch, an observer prescription and null-ray propagation. Possible deviations from the Schwarzschild shadow therefore remain model-dependent research targets.

12 Observable Implications

The TIG transition may affect observables sensitive to horizon-scale geometry. Possible implications include:

- modifications of black-hole shadow radius;
- altered photon-sphere structure;
- changes in near-horizon lensing behavior;
- modified ringdown or echo-like structures near criticality.

At the present stage, these implications are model-dependent research targets. A prediction requires specified sources, propagation and perturbation equations, and a demonstrated map from the geometry to the observable. The horizon-root fold alone does not establish enhanced sensitivity or a critical exponent for every proposed observable.

13 Literature Context and Theorem Boundaries

Static horizon roots and dynamical formation are distinct. Hayward’s time-dependent example supplies mass evolution and fluxes [13]; these are additional inputs, not consequences of our cubic. Trapping horizons need not be global event horizons.

The zeroth-law analysis of Ghosh and Sarkar [15] assumes a smooth stationary Killing horizon and appropriate field equations and matter conditions. Its Ricci-squared argument uses nonzero coupling, a compact connected boundaryless section and non-extremality. Pure $f(R)$ uses the mixed projection and requires $f_R \neq 0$ instead. Our critical horizon has $\kappa = 0$; spherical angular constancy does not prove a field-equation solution or horizon formation.

Bessa [16] studies scalar dust perturbations of flat FLRW in full quadratic gravity, including Weyl squared, with sub-horizon and quasi-static approximations. Its formal infrared continuation does not prove TIG perturbative stability or a controlled super-horizon limit.

Garay et al. [17] provide an indirect two-vertex LQG comparison, not ordinary quantum mechanics recovered from TIG. Finite-dimensional quantum reconstruction [18] requires its own operational assumptions. No TIG derivation of these assumptions or quantum recovery is supplied here.

14 Conditional Round-Trip Delay

In a connected static region with $F > 0$, radial null propagation gives $dt = \pm dr/F$. We additionally prescribe an inner reflector at $r_{\text{ref}} = 2Mx_{\text{ref}} > 0$ and a partially reflecting, transmitting outer scattering boundary at $r_b = 2Mx_b$. Neither boundary is derived from TIG. The geometric round-trip time in asymptotically normalized t is

$$T_{\text{rt}} = 2 \int_{r_{\text{ref}}}^{r_b} \frac{dr}{F} = 4M \int_{x_{\text{ref}}}^{x_b} \frac{x^3 + \beta^3}{x^3 - x^2 + \beta^3} dx. \quad (46)$$

Echo-like output additionally requires excitation and appropriate scattering; this formula supplies neither a waveform nor a gravitational perturbation equation. A minimally coupled massless test scalar would have

$$(-\partial_t^2 + \partial_{r_*}^2 - V_\ell)\psi_\ell = 0, \quad \frac{dr_*}{dr} = \frac{1}{F}, \quad V_\ell = F \left[\frac{\ell(\ell+1)}{r^2} + \frac{F'}{r} \right]. \quad (47)$$

An ideal Dirichlet inner boundary reflects that test field; its physical origin and the required outer scattering remain assumptions. The horizonless-object calculation of Cardoso et al. [14] is a methodological comparison with different geometry and boundary data. Its one-way delay is distinct from the round trip defined here.

For fixed M , set $\eta = \beta - \beta_c > 0$, $\delta = x - x_c$, $D_0 = 4/9$ and $s = \sqrt{3}\beta_c\sqrt{\eta}$. Above criticality $F > 0$ at all positive radii. The exact numerator is

$$P = \delta^2 + \delta^3 + 3\beta_c^2\eta + 3\beta_c\eta^2 + \eta^3. \quad (48)$$

Its bottleneck has $F \sim (\delta^2 + s^2)/D_0$. If the fixed endpoints satisfy $0 < x_{\text{ref}} < x_c < x_b$, the scaled integral in Eq. (46) contains $\int_{-\infty}^{\infty} dy/(1+y^2) = \pi$ and hence

$$T_{\text{rt}} \sim \frac{16\pi M}{9\sqrt{3}\beta_c}(\beta - \beta_c)^{-1/2}, \quad \beta \downarrow \beta_c, \quad \beta > \beta_c. \quad (49)$$

The ratio to this leading term tends to one. The local scaled integrand is dominated by a constant times $(1+y^2)^{-1}$; regions away from x_c have bounded travel time. One fixed endpoint at x_c halves the coefficient; two fixed endpoints strictly on the same side have a finite limit. A fixed exterior observer with a finite positive limiting lapse measures $\Delta\tau_{\text{obs}} = \sqrt{F(r_{\text{obs}})}T_{\text{rt}}$.

Below criticality, let $\epsilon = \beta_c - \beta > 0$ and $s = \sqrt{3}\beta_c\sqrt{\epsilon}$. Returning exterior paths must stay at $x > x_+$. Fixed $x_c < x_{\text{ref}} < x_b$ give a finite critical limit. Only the additional tracking prescription $x_{\text{ref}} = x_+ + (q-1)s$, with fixed $q > 1$, gives

$$T_{\text{rt}} \sim \frac{2MD_0}{\sqrt{3}\beta_c} \ln \frac{q+1}{q-1} \epsilon^{-1/2}. \quad (50)$$

This prescribes a coordinate gap, not a fixed proper gap. At fixed subcritical β , a mirror approaching the simple outer horizon instead gives $T_{\text{rt}} = -\kappa_+^{-1} \ln(h/L) + O(1)$, where $h = r_{\text{ref}} - r_+ \rightarrow 0^+$ and $L > 0$ is fixed. At exact criticality, $x_{\text{ref}} = x_c + d$ gives $T_{\text{rt}} \sim 16M/(9d)$ as $d \rightarrow 0^+$. Crossing a future outer horizon is not a returning exterior signal path. The former unrestricted $(\beta_c - \beta)^{-1/2}$ echo claim therefore does not hold for arbitrary boundaries. Equation (49) is a conditional geometric delay, not a demonstrated physical TIG echo.

15 Scope and Limitations

This paper develops the algebraic horizon structure of TIG. Several important questions remain open.

First, the complete dynamical field equations associated with the TIG consistency principle must be formulated. Second, stability of the resulting compact configurations must be analyzed. Third, observational predictions must be computed within a fully specified metric model.

The present work is an algebraic analysis of a specified representative geometry with a local fold transition. It is not a complete gravitational theory, a vacuum solution of the tested quadratic metric action, or a proof of its own observational predictions.

16 Conclusion

For the specified regular mass profile, we derived the exact horizon equation

$$x^3 - x^2 + \beta^3 = 0, \quad (51)$$

with critical point $x_c = 2/3$, $\beta_c = (4/27)^{1/3}$. Two positive roots coalesce in a local nondegenerate fold and disappear above the critical value; their displacements scale as $(\beta_c - \beta)^{1/2}$.

The cubic degree and the correction parameterization follow from the selected mass profile, not from a universal cubic requirement for folds. The geometry has a Schwarzschild limit and a finite-curvature core, but fails the vacuum trace equation of the tested constant-coefficient quadratic metric action. A dynamical completion with specified sources, perturbative stability, and observable predictions must be established separately.

The static family is identified with Hayward's earlier metric. With additionally prescribed reflection and fixed endpoints straddling the horizonless bottleneck, the round-trip delay has exponent $-1/2$ on the $\beta > \beta_c$ side. Subcritical delays depend on the mirror prescription. Neither this kinematic calculation nor the comparison literature establishes physical TIG echoes, dynamical horizon formation or quantum-mechanical recovery.

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